PPP-T-76C
May 10, 1976
SUPERSEDING
Fed. Spec. PPP-T-76B
July 16, 1963

FEDERAL SPECIFICATION

TAPE, PACKAGING, PAPER (FOR CARTON SEALING)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers a water resistant, pressure-sensitive adhesive tape of one type, for use in closing and sealing fiberboard containers.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-T-680 - Tape, Pressure Sensitive Adhesive: Packaging and Packing Of.

Federal Standard:

Fed. Test Method Std. No. 147 - Tapes, Packaging, Adhesive and Gummed: Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks at the Index of Federal Specifications and Standards from established distribution points in their agencies.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:

D 2860 - Adhesion of Pressure-Sensitive Tape to Fiberboard at 90-Degree Angle and Constant Stress.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

FSC 8135

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3. REQUIREMENTS

3.1 Construction. The tape shall have a paper backing which has been treated to resist water. This backing shall be coated on one side with homogeneous water-resistant, pressure-sensitive adhesive applied in a smooth layer. The tape shall be able to adequately perform over the temperature range of -54° to 71°C (-65° to 160°F), and shall require no solvent, heat or other preparation prior to application.

3.2 Finished tape.

3.2.1 Rolls. The tape shall be uniformly and evenly wound in rolls, adhesive side in, on cores of sufficient rigidity to prevent distortion of the rolls under normal conditions of transportation, storage, and use. The inside diameter of the core shall be 3 -0 + 1/16 inches. No more than four splices will be permitted in any roll. All splices shall be such that they will not separate when the roll is unwound in test or application.

3.2.2 Length and width. The width of the tape shall be 1, 1-1/2, 2, or 3 inches as specified in the contract or order (see 6.2). A tolerance of + 1/32 inch shall be allowed on all widths. The length shall be evenly wound in 120 yard rolls.

3.3 Physical properties. The tape shall meet the requirements listed in table I when tested as described in Section 4.

TABLE I

	Minimum Value (unless noted)	Paragraph
Adhesion, oz./in.:		
At 23°C (73°F)		
As received	35	4.3.6
After accelerated aging	35	4.3.7
After accelerated weathering	35	4.3.8
At 10°C (50°F)		
As received	10	4.3.6
After accelerated aging	10	4.3.7
At -54°C (-65°F)	-	4.3.5
To own backing		
As received	10	4.3.6
After accelerated aging	10	4.3.7
Adhesion to Fiberboard at 90° and Constant Stress, minutes:	30	4.3.9
Curling, inches:		
As received	10 max.	4.3.4
After accelerated aging	10 max.	4.3.10
Holding Power, hours:		
At 23°C (73°F)		
As received	96	4.3.11.1
After accelerated aging	96	4.3.11.1
At 70°C (158°F)		
As received	96	4.3.11.2
After accelerated aging	96	4.3.11.2
Roll Distortion, inches:		
Radius	0.5 max.	4.3.12
Width	0.5 max.	4.3.12
Tearing Resistance, weaker direction, grams:		
As received	100	4.3.13
After accelerated aging	75	4.3.13
After accelerated weathering	75	4.3.13

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TABLE I (Cont.)

	Minimum Value (unless noted)	Paragraph
Tensile Breaking Strength, lbs./in. width:		
As received:		
Dry longitudinal	45	4.3.14.1
Dry transverse	22	4.3.14.1
Wet longitudinal	20	4.3.14.2
Wet transverse	12	4.3.14.2
After accelerated aging.		
Dry longitudinal	40	4.3.14.1
Dry transverse	20	4.3.14.1
Thickness, inch	0.010 max.	4.3.4
Twisting, degrees:		
As received	360 max.	4.3.4
After accelerated aging	360 max.	4.3.10
Unwinding Properties:		
Resistance to unwinding, lbs./in. width:		
As received	3 max.	4.3.15
After accelerated aging	3 max.	4.3.15
Water Solubility, percent	2.0 max.	4.3.4

3.4 Workmanship. The tape shall be free of defects which detract from its appearance or impair its serviceability. The adhesive shall be a smooth uniform coating, covering the entire area of one side of the tape. Edges shall be straight, true, and unbroken.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling and examination.

4.2.1 End item examination. Sampling and examination of the item shall be as specified in Section 7 of Fed. Test Method Std. No. 147.

4.2.2 End item testing. Sampling and examination for end item testing shall be as specified in Section 7 of Fed. Test Method Std. No. 147 at an Acceptable Quality Level (AQL) of 4 percent.

4.2.3 Examination of preparation for delivery. The examination of preparation for delivery shall be performed in accordance with PPP-T-680.

4.3 Tests.

4.3.1 Preparation for testing. The specimen rolls shall be selected and conditioned as specified in Method 4 of Fed. Test Method Std. No. 147.

4.3.2 Accelerated aging. All specimens selected for testing of aged material shall be aged as described in Method 62 of Fed. Test Method Std. No. 147 at a temperature of 57° ± 1°C (135° ± 2°F) for 20 consecutive days. After aging, specimens shall be conditioned in accordance with 4.3.1.

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4.3.3 Accelerated weathering. All specimens selected for testing of weathered material shall be exposed in a weatherometer as specified in Method 63 of Fed. Test Method Std. No. 147 for a period of 60 hours. After weathering, specimens shall be conditioned in accordance with 4.3.1.

4.3.4 Test procedures. The following tests and methods used shall be conducted according to Fed. Test Method Std. No. 147:

Test	Method No.
Curling and twisting	84
Thickness	36
Water solubility	74

4.3.5 Adhesion at -54°C (-65°F). Specimens shall be tested as described in Method 14 of Fed. Test Method Std. No. 147. The tape shall show no evidence of release from the kraft paper, nor shall there be release of adhesive from the backing of the tape.

4.3.6 Adhesion, as received. Specimens shall be tested for adhesion as described in Method 10 of Fed. Test Method Std. No. 147 at 23°C (73°F), 10°C (50°F), and to its own backing.

4.3.7 Adhesion after accelerated aging. Specimens shall be aged as specified in 4.3.2. After aging, the tape shall be tested for adhesion as described in Method 10 of Fed. Test Method Std. No. 147 at 23°C (73°F), 10°C (50°F), and to its own backing.

4.3.8 Adhesion after accelerated weathering. Specimens shall be exposed in a weatherometer as specified in 4.3.3. After accelerated weathering, the tape shall be tested for adhesion as described in Method 10 of Fed. Test Method Std. No. 147 at 23°C (73°F).

4.3.9 Adhesion to fiberboard at 90° and constant stress. The tape shall be tested in accordance with ASTM Method D 2660, Procedure A. The linerboard used as the standard test surface shall be Standard Reference Material 1810, which is available from the Office of Standard Reference Materials, National Bureau of Standards, Washington, DC 20234. The tape shall adhere to the linerboard for a minimum of 30 minutes.

4.3.10 Curling and twisting after accelerated aging. Specimens shall be aged as specified in 4.3.2. After aging, the tape shall be tested for curling and twisting as described in Method 84 of Fed. Test Method Std. No. 147.

4.3.11 Holding power.

4.3.11.1 At 23°C (73 ± 2°F). Specimens from tape as received and after aging (see 4.3.2) shall be tested as described in Method 20 of Fed. Test Method Std. No. 147. A weight of 1000 ± 5 grams shall be allowed to act uniformly upon the specimens for 96 hours at 23°C (73 ± 2°F). The tape shall show no more than 1/8 inch creepage or slippage on the panel within 96 hours, as indicated in table I.

4.3.11.2 At 70° ± 1°C (158° ± 2°F). Specimens shall be prepared and tested as specified in 4.3.11.1 except that a 100 ± 0.5 gram load shall be applied and allowed to act for 96 hours in a circulating air cabinet at 70° ± 1°C (158° ± 2°F). The tape shall show no more than 1/8 inch creepage or slippage on the panel within 96 hours as indicated in table I.

4.3.12 Roll distortion. Sample rolls shall be placed flat on the side on a sheet of paper; the circumference shall be traced and the greatest radius measured to the nearest 1/32 of an inch. The width shall be determined to the same accuracy by measuring the distance between the base of the roll and the top of the uppermost ply of tape. After aging in accordance with 4.3.2, the above measurements shall again be made. The individual roll distortions shall be averaged and shall be reported to the nearest 1/32 of an inch.

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4.3.13 Tearing resistance. Tearing resistance shall be determined on material as received, after aging (see 4.3.2) and after weathering (see 4.3.3), in accordance with Method 40 of Fed. Test Method Std. No. 147.

4.3.14 Tensile strength.

4.3.14.1 Dry longitudinal and transverse. Dry longitudinal tensile breaking strength shall be conducted in accordance with Method 30 of Fed. Test Method Std, No. 147 on all width tapes both as received and after accelerated aging (see 4.3.2). Transverse tensile breaking strength shall be conducted only on rolls of tape two inches or more in width.

4.3.14.2 Wet longitudinal and transverse. Specimens shall be immersed in distilled water at 23° ± 1°C (73° ± 2°F) for 4 hours, then tested as specified in 4.3.14.1 except that only material as received shall be tested for wet tensile breaking strength.

4.3.15 Unwind. Resistance to unwind shall be determined on material as received and after accelerated aging (see 4.3.2), in accordance with Method 80 of Fed. Test Method Std. No. 147. There shall be no adhesive transfer, nor tearing or delamination of the backing.

5 PREPARATION FOR DELIVERY

5.1 Packaging and packing. Packaging and packing shall be in accordance with the applicable requirements of PPP-T-680. The level of packaging and packing shall be A, B, or C, as specified in the contract or order (see 6.2).

5.2 Marking. Containers of tape shall be marked in accordance with the requirements of PPP-T-680. When specified (see 6.2), the core of each roll of tape shall be marked on the inside or on the edge with numerals or letters indicating the month and year of manufacture and specification number. In addition, the manufacturer's name and designation of the product shall be marked on the core.

6. NOTES

6.1 Intended uses. This tape is intended for applications where a water-resistant tape is required to close and seal fiberboard containers.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Width of tape required (see 3.2.1).
(c) Selection of applicable levels of packaging and packing (see 5.1).
(d) Marking required (see 5.2).

6.3 Storage. The tape should be stored in the original container in a cool location. It should not be stored in close proximity to steam pipes, radiators, or other sources of heat, or in a location where it may be exposed to direct rays of the sun.

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MILITARY INTERESTS:

Custodians:

Army - GL
Navy - AS
Air Force - 69

Review Activities:

Army - MC, MI, EL
Navy - SA
Air Force - 71, 80, 84

CIVIL AGENCY COORDINATING ACTIVITIES:

DCG
ERDA
GPO
HEW-HSM
NASA-LRC
NBS
VA-DMS

PREPARING ACTIVITY:

GSA-PBS

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